

4.4. Reformulation of the technical problem

In accordance with the case law of the boards, the technical problem can be reformulated, and in certain circumstances actually has to be reformulated, since the only factor of importance in determining the problem objectively is the result actually achieved in relation to the closest state of the art (see e.g. T.1397/08, T.2406/18). The problem as first formulated can be modified even at the appeal stage (see e.g. T.162/86, OJ 1988, 452, T.1397/08; T.2371/13; T.659/15) as long as the spirit of the original disclosure of the invention is respected. See also T.2371/13, T.659/15. However, a reformulation of the technical problem is conditioned by certain limitations (see in this chapter I.D.4.4.3).

4.4.1 Reformulation if the problem is not solved

The technical problem may have to be reformulated, in particular in **less ambitious terms**, if the combination of features in the claim does not solve this problem over the whole area defined in the claim (T.39/93, OJ 1997, 134; T.400/98; T.235/04; T.96/06; T.1159/12; T.2001/12; T.1279/14; T.44/17).

In T.87/08 the board stated that Art. 56 EPC required that the assessment of inventive step was made "having regard to the state of the art". The board stated it was established in the case law of the boards of appeal that a decision was not sufficiently reasoned in the sense of R. 68(2) EPC 1973 (R. 111(2) EPC) if the chain of reasoning to justify the finding of lack of inventive step merely stated that a purported effect had not been achieved, i.e. this technical problem had not been solved, without reformulating the problem in a less ambitious way and without assessing obviousness of the claimed solution to that reformulated problem in the light of the cited prior art. See also T.306/09, T.2375/10, T.1212/11, T.2186/11.

In T.143/13 the board observed that, according to the problem and solution approach, it was the problem described in the patent application that was normally used as the starting point for assessing inventive step. It noted that in the appealed decision, the step of assessing whether the problem defined in the application had been solved had been omitted. Nevertheless, the problem had been reformulated in terms that had not been invoked by the appellant (applicant). The justification provided for the reformulation of the problem was that the originally defined problem was artificial. However, the board could not see why and considered the reformulation of the problem in the appealed decision was not appropriate.

4.4.2 Reformulation of the "subjective" into an objective technical problem

According to T.39/93 (OJ 1997, 134), although it is desirable to take the same approach as the applicant regarding the definition of the technical problem (see in this chapter I.D.4.2.2), the technical problem as originally presented in the application or patent, which was to be regarded as the "subjective" technical problem, might require reformulation on the basis of objectively more relevant elements originally not taken into account by the applicant or patentee. This reformulation yields a definition of the "objective" technical

problem. The latter represented the ultimate residue (effect), corresponding to the objective contribution provided by the subject-matter defined in the relevant claim.

It is established case law of the boards of appeal that where a specific problem is identified in the description, the applicant or patentee may be allowed to put forward a modified version of the problem particularly if the issue of inventiveness has to be considered on an objective basis against a new prior art which comes closer to the invention than that considered in the original application or granted patent specification (see e.g. T 386/89; T 39/93, OJ 1997, 134; T 165/05; T 716/07).

As a matter of principle, any effect provided by the invention may be used as a basis for reformulating the technical problem, as long as that effect is derivable from the application as filed (T 452/05, T 698/08, T 188/09, T 2483/11, T 605/14, T 1196/12, T 1406/15). A reformulation of the problem also may be appropriate if an alleged effect of a described feature could be deduced by the skilled person from the application in the light of the prior art or if new effects submitted subsequently during the proceedings were implied by or related to the technical problem initially suggested. In relation to new effects it was not permissible to change the nature of the invention (T 13/84, OJ 1986, 253; T 344/89; T 767/02; T 698/08).

4.4.3 Limitations to the reformulation of the technical problem

a) General principles and relationship with Article 123(2) EPC

As has been established in the case law of the boards of appeal (see in this chapter at I.D.4.4.2), any technical effect provided by the invention may be used as a basis for reformulating the technical problem, as long as that effect is derivable from the application as filed.

A reformulation of the problem is thus not precluded by Art. 123(2) EPC 1973 if the problem can be deduced by the skilled person from the application as filed when considered in the light of the closest prior art (T 13/84, OJ 1986, 253; T 469/90; T 530/90; T 547/90; T 39/93, OJ 1997, 134; T 375/93; T 687/94; T 877/06).

In T 818/93 the board added that it sufficed if the reformulated problem could be deduced later by comparing the application with the closest prior art. Since features from the drawings might be incorporated into the claims, and also into the description in support of the claims (referring to T 169/83, OJ 1985, 193), those features' effects and advantages might also be used as a basis for reformulating the problem, provided this problem could clearly be deduced from the above comparison.

In T 184/82 (OJ 1984, 261) doubt was cast on the general validity of the statements in the specification and examples (hot bacon test) which suggested that the specific problem had been successfully solved by the invention. Nevertheless, the board found the person skilled in the relevant art would have recognised that the original problem was not unrelated to the more general aim of improving the heat and solvent resistance of materials for containers. The board stated in its headnote that technical success at a general level

could replace failure at a more specific level when assessing the effect of the invention provided the skilled person could recognise the same as implied or related to the problem initially suggested. See also T 732/89, T 106/91, T 767/02.

In T 1422/12 claim 1 related to crystalline forms of tigecycline. The formulation of the technical problem to be solved fell well within the framework of the invention as disclosed in the application in suit. The board noted that any effects may be taken into account, so long as they concerned the same field of use and do not change the character of the invention (see T 440/91). That a more specific problem of improved stability with respect to epimerisation was not mentioned in the application as originally filed was irrelevant since improvement of stability by avoidance of epimerisation, and, as a consequence, improved biological activity was clearly recognisable by the skilled person as a desirable effect for a tetracycline antibiotic from the application as originally filed (see also T 39/93, OJ 1997, 134).

In T 564/89 the appellant (opponent) submitted that any amendment of the technical problem had to be in line with Art. 123(2) EPC 1973. The board stated that this article was not concerned with the issue of whether or not an objectively reformulated technical problem could be used in the course of the so-called problem and solution approach. Art. 123(2) EPC 1973 would only come into play if an amended technical problem were incorporated into the description itself. See also T 284/98, T 276/06.

b) Subsequently invoked technical effect

As a rule, additional advantages not mentioned in the application as originally filed but relating to a mentioned field of use can also be considered for the purpose of assessing inventive step, provided such advantages do not alter the character of the invention (T 440/91; see also T 1062/93, T 1983/07, T 1422/12, T 1450/14, T 321/16). The character of the invention is not altered if the technical problem specified in the application as filed is supplemented by such advantages, since the skilled person could consider them on account of their close technical relationship to the original problem (T 440/91, T 1062/93, T 1983/07, T 1450/14, T 321/16).

On the other hand, an alleged technical effect of a described feature cannot be taken into account when determining the problem underlying the invention for the purpose of assessing inventive step, if it cannot be deduced by the skilled person from the application as filed considered in relation to the nearest prior art (T 386/89), or if it is not at least hinted at in the originally filed documents (T 321/16; in T 344/89 "foreshadowed in the originally filed application"). See also T 440/91, T 532/00, T 867/13.

In T 1188/00 the board found that a reformulation relating to an effect first alleged in appeal proceedings (a more ambitious problem) could not be used to substantiate inventive step unless it was credibly demonstrated that the alleged effect could be achieved across the whole scope of the claim. The burden of proof for this lay with the patent proprietor. See also T 134/00, T 815/16 (referred to in terms of plausibility in T 253/02, T 172/07, T 366/12, T 438/16, T 815/16, T 987/17).

In T.235/04 the board did not accept the formulation of the technical problem since the purported improvement had not been demonstrated to arise within the whole area that was claimed. The board explained that when defining the technical problem, an effect could not be retained if the promised result was not attainable throughout the entire range covered by the claimed subject matter. Therefore, the technical problem needed to be redefined in a less ambitious way (T.626/90, T.1057/04, T.824/07).

In T.259/05 too the board held the claimed improvement to be unproven; it had not been substantiated by tests or plausibly shown in any other way. The ambitious problem formulated could therefore not be considered successfully solved. As a result, the claimed improvement in the method's efficiency could not be taken into account for assessing the objective problem which the patent in suit sought to solve, or the inventiveness involved. The technical problem therefore had to be reformulated less ambitiously.

According to T.155/85 (OJ 1988, 87), it was not acceptable to rely on an effect which had previously been described as undesirable and of no value by the applicant, to present it suddenly as possibly representing an advantage from another point of view, and thereby to imply that the technical problem and the considerations of inventive step should take this reversal into account. A redefinition of the technical problem should not contradict earlier statements in the application about the general purpose and character of the invention. See also T.115/89, T.2245/10.

4.5. Alternative solution to a known problem

Art. 56 EPC does not require that the problem to be solved should be novel in itself. The fact that the underlying problem of the patent have already been solved by the prior art does not necessarily require redefinition of the problem for the assessment of inventive step, if the subject-matter of the patent represents an alternative solution to this problem. (T.92/92, with reference to T.495/91; see also T.1074/93, T.780/94, T.323/03 T.78/05).

For an inventive step to be present, it is not necessary to show improvement – substantial or gradual – over the prior art (T.100/90, T.588/93, T.620/99 T.1791/08). An earlier solution to a given technical problem does not preclude later attempts to solve the same problem in another, non-obvious way (T.615/05). In T.2081/15 the non-obvious alternative solution was "technically plausible" and involved an inventive step, albeit a small one, over D1 and the skilled person's common general knowledge.

On the other hand, in T.1179/16 the board noted that if the only contribution of the invention was to propose something different from the prior art (i.e. the provision of an alternative), then it was usually appropriate to consider that the skilled reader would take into account any alternative known in the underlying technical field (unless the closest prior art teaches away from it). The board stated that in such cases it might not be required to justify the selection of a particular solution, because it was assumed that an invention based on incorporating known features for the sole purpose of establishing novelty must be rendered obvious by a corresponding step of selecting any alternative known in the art.

In T.144/16 the board found the technical problem needed to be reformulated in a less ambitious way, and in view of the teaching of document (1), as the provision of alternative adhesive compositions. It noted that when looking for alternative compositions, the skilled person does not restrict the teaching of document (1) to its preferred embodiments, but takes into consideration all features taught in that document. In T.148/10 the board stated that the skilled person required no inventive skill to make a selection from known alternatives. The skilled person was able to consider advantages and disadvantages of their selection and would weigh them against each other.

5. "Could-would approach"

To determine whether the claimed invention, starting from the closest prior art and the objective technical problem, would have been obvious to the skilled person, the boards apply the **"could-would approach"** (see also Guidelines G-VII, 5.3 – March 2022 version). This means asking not whether the skilled person **could** have carried out the invention, but whether he **would have done** so in the expectation of solving the underlying technical problem or in the expectation of some improvement or advantage (T.2/83, OJ 1984, 265; T.90/84; T.7/86, OJ 1988, 381; T.200/94, T.885/97, T.1148/15). When considering whether or not claimed subject-matter constitutes an obvious solution to an objective technical problem, the question to be answered is whether or not the skilled person, in the expectation of solving the problem, would have modified the teaching in the closest prior art document in the light of other teachings in the prior art so as to arrive at the claimed invention (T.1014/07, T.867/13). So the point is not whether the skilled person could have arrived at the invention by modifying the prior art, but rather whether, in expectation of the advantages actually achieved (i.e. in the light of the technical problem addressed), he would have done so because of promptings in the prior art (T.219/87, T.455/94, T.414/98).

In T.1014/07 the board stated that since this question involved determining whether or not the skilled person would – and not could – have made a particular modification, it was necessary for answering the question to identify conclusive reasons on the basis of tangible evidence that would have prompted the skilled person to act in one way or the other. The board in T.1045/12 found that its decision, being based on prior-art documents D4 and D3, was based on "tangible evidence" (referring to T.1014/07).

It has been held that once an invention existed, it could often be shown that the skilled person could have made it by combining different elements in the prior art, but such arguments had to be disregarded as the product of ex post facto analysis (T.564/89).

According to T.939/92 (OJ 1996, 309), the answer to the question what a skilled person would have done depended in large measure on the technical result he had set out to achieve. In other words, the notional "person skilled in the art" was assumed to act not out of idle curiosity but rather with a specific technical purpose in mind (see also T.28/20).

In T.867/13 the board pointed out that what the skilled person, starting from the closest prior art and faced with the objective technical problem, would or would not do depends